METRIC

MIL-PRF-85045/24B(SH)
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23 August 2022

USED IN LIEU OF
MIL-PRF-85045/24B
5 November 2021

PERFORMANCE SPECIFICATION SHEET

CABLE, FIBER OPTIC, NINETY FIBERS, ENHANCED PERFORMANCE, CABLE CONFIGURATION TYPE 2 (OFCC), APPLICATION B (SHIPBOARD), CABLE CLASS SM AND MM

This specification is approved for interim use by the Naval Sea Systems Command. Other activities in the Department of Defense may use this interim amendment or may continue using MIL-PRF-85045/24B.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-85045.

CLASSIFICATION:

Fiber optic cable configuration type: 2 (optical fiber cable component [OFCC])

Fiber optic cable class: MM (graded-index, glass core and glass cladding, multimode)

SM (dispersion-unshifted, glass core and glass cladding, single-mode)

DESIGN AND CONSTRUCTION:

Fiber: Class MM fibers shall be in accordance with MIL-PRF-49291/6.

Class SM fibers shall be in accordance with MIL-PRF-49291/7.

Buffer diameter: 900±50 micrometers (µm)

OFCC:

Dimensions and configuration: See figure 1.

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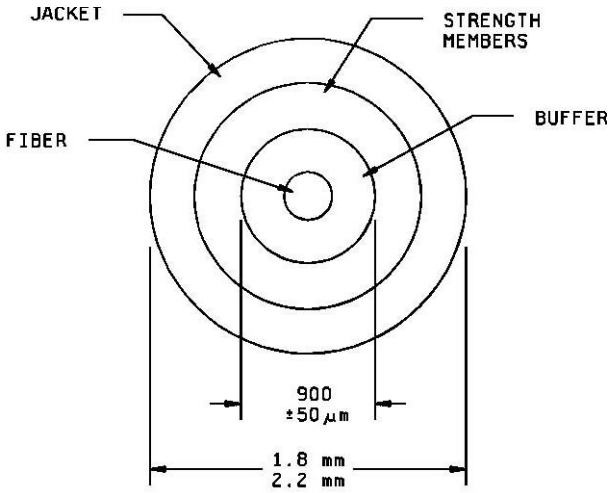


FIGURE 1. OFCC.

Mass per unit length: <15 kilograms per kilometer (kg/km)

Tensile loading: >270 newtons (N)

Dynamic bend tensile load: 90 N minimum

Jacket material: The OFCC jacket shall be composed of a low-halogen, low-smoke, low-toxicity polymer material.

Cable jacket color: Shall be black or blue.

OFCC color:

Slate (MIL-PRF-49291/6 fiber)

Yellow (MIL-PRF-49291/7 fiber)

OFCC marking: Each OFCC shall be uniquely marked with a number between 1 and 18. The form of the marking shall be the printed spelling of the number, followed by a dash, followed by the printed Arabic numeral. The marking shall be applied and repeated every 0.10 meter (m) along the OFCC jacket.

Short-term minimum bend diameter: Eight times the OFCC outer diameter.

Long-term minimum bend diameter: Sixteen times the OFCC outer diameter.

Eighteen-fiber sub-cables:

Eighteen-fiber sub-cables shall be in accordance with M85045/22-01 or M85045/22-02 standard performance, except that the minimum outer jacket thickness shall be not less than 0.9 millimeter (mm).

OFCC 1 shall be located in the innermost ring of OFCCs. OFCCs shall be consecutively numbered starting with those in the innermost ring (1 through 6) and then those in the outer ring (7 through 18). OFCCs 1 and 7 shall be located along approximately the same radial line within the cable.

Sub-cable marking: Each sub-cable shall be uniquely marked with a sub-cable number between 1 and 5. The marking shall be applied and repeated every 0.5+0.1 m along the sub-cable jacket. The form of the marking shall be the printed spelling of the sub-cable number, followed by a dash, followed by the printed Arabic sub-cable numeral.

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Finished cable:

Dimensions and configuration: See figure 2. Five sub-cables shall be helically laid over the central member. The minimum outer jacket thickness shall be not less than 1.1 mm and shall be determined from the measurements made at the high point of each sub-cable, five in total. The minimum average jacket thickness shall be not less than 1.5 mm and shall be determined from the measurements made at the high point of each sub-cable, five in total.

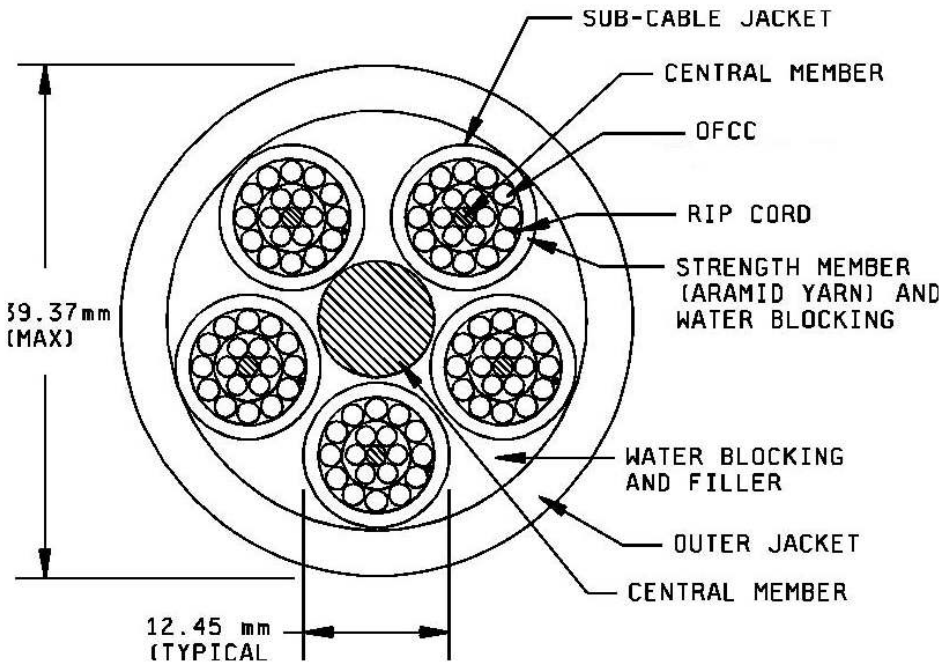


FIGURE 2. Ninety-OFCC fiber optic cable.

Number of fibers: 90 (one per OFCC)

Concentricity: See Dimensions and configuration.

Mass per unit length: ≤1,450 kg/km

Jacket material: The overall jacket shall be composed of a low-halogen, low-smoke, low-toxicity polymer material.

Short-term minimum bend diameter: Eight times the cable outer diameter. (The short-term minimum bend diameter shall be used in all environmental and mechanical tests that specify a cable minimum bend diameter.)

Long-term minimum bend diameter: Sixteen times the cable outer diameter.

Minimum continuous length: The minimum continuous length of all cables shall be not less than 0.5 km. If lengths less than 0.5 km are specified in the purchase order, a conformance inspection shall be performed on test specimens not less than 0.5 km in length from which the purchase order lengths are cut.

PERFORMANCE REQUIREMENTS: The individual performance requirements shall be as specified in MIL-PRF-85045 and this specification sheet, including the requirements identified in table I.

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Optical properties:

Maximum attenuation rate: 4.5 decibels (dB)/km at 850±20 nanometers (nm), 2.0 dB/km at 1,300±20 nm for class MM fiber.

2.0 dB/km at 1,310±20 nm and 1,550±20 nm for class SM fiber.

For cables with radiation cross-linked jackets, the change in attenuation rate measurement may be made up to 30 days after cross-linking of the cable jacket.

Bandwidth: Fiber with a minimum bandwidth of 500 megahertz (MHz)-km at 1,300 nm and 160 MHz-km at 850 nm shall be used (multimode cables only).

Change in optical transmittance: Measurements shall be made at 1,300±20 nm. A minimum of 15 fibers shall be monitored for change in optical transmittance. A minimum of one fiber shall be monitored in the inner ring of OFCCs of each sub-cable. A minimum of two fibers shall be monitored in the outer ring of OFCCs of each sub-cable. The fibers monitored in each ring shall be randomly selected, but different fibers shall be monitored in each test sample. For shock testing, only 4 fibers are required to be monitored (1 each in 4 of the 5 18-fiber sub-cables).

Mechanical properties:

Low temperature flexibility: The exposure temperature shall be -40 °C.

Cyclic flexing: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same test position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Cable twist-bending: 500 cycles at 25±2 °C and 100 cycles at -28±2 °C. Change in optical transmittance measurements shall be made every 100 cycles for the 500-cycle exposure and every 25 cycles for the 100-cycle exposure. Each change in optical transmittance measurement shall be performed with the test specimen in the same test position in the test cycle. The cycling may be halted to perform the change in optical transmittance measurement.

Hosing: Both low pressure and hydrostatic pressure are applicable.

Low pressure: Leakage shall be not greater than 450 milliliters (mL).

Hydrostatic: 2.1 megapascals (MPa). A hydrostatic test shall be performed on the cable and sub-cable. For the cable, the stuffing tube may be torqued to 115 newton-meters (N m). For the sub-cable, the stuffing tube shall be torqued to 41 N m.

Cable scraping resistance: 250 cycles. The length from the top center of the rotating cylinder to the end of the cable specimen shall be between 762 mm and 788 mm (30 to 31 inches). The test weight shall not be used. The specimen shall be oriented so that the abrading element contacts the cable at the minimum cable jacket wall location. This minimum cable jacket wall location will coincide with the high point of a sub-cable.

Cable-to-cable abrasion: 500 cycles

Stuffing tube compression: A fully assembled cable specimen shall be tested as follows. One end of the specimen shall be fitted into the appropriate size of bulkhead stuffing tube (refer to MIL-STD-2042 for guidance in choosing the appropriate size bulkhead stuffing tube). The stuffing tube shall be torqued to 115 N m (the maximum distance between the gland nut and the stuffing tube body shall be 3.2 mm) and conditioned at ambient temperature for 24 hours. The stuffing tube shall be torqued a second time to 115 N m, conditioned for a second 24 hours, torqued a third time, and conditioned for a final 24 hours. Change in optical transmittance measurements shall be made approximately 5 minutes before and after the stuffing tube is torqued and once every 3 hours during the conditioning periods. The change in optical transmittance shall be not greater than 0.5 dB during and after the test.

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Environmental properties:

Temperature range:

Operating:	-28 to 65 °C
Nonoperating:	-40 to 70 °C
Storage:	-40 to 70 °C

Temperature cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made over a period of 1 hour at the end of each temperature plateau.

Temperature humidity cycling: Change in optical transmittance measurements may be made periodically. At a minimum, one optical transmittance measurement shall be made at the end of each temperature plateau.

Temperature life (life aging): The jacket material shall be tested at 175 °C for 4 hours.

Fluid immersion: Exposure to automobile gasoline and tap water are not required and the following test temperatures shall be used for the fluids indicated: fuel oil (98 to 100 °C), turbine fuel (48 to 50 °C), and lubricating oil (98 to 100 °C).

Chemical properties:

Halogen content: <0.2 percent

Cross-link verification: This test is applicable for cables with cross-linked jackets only. The test shall be conducted in accordance with ICEA T-28-562 and run at 200 °C. The test shall be sequenced after the weathering test in the qualification test sequence and after the fluid immersion test in the group C conformance test sequences. The hot creep shall not exceed 100 percent, and the hot creep set shall not exceed 10 percent.

Part or identifying number (PIN):

M85045/24-01 (Multimode)

M85045/24-02 (Single-mode)

Qualification and conformance inspection: See table I. This table specifies applicable requirements, qualification inspections, and conformance inspections per this specification sheet. The “Requirement paragraph” and “Test paragraph” columns within this table reference MIL-PRF-85045. The “Cable length” column of this table applies only to the qualification inspection. A reduced sample set is allowed for conformance inspection; refer to MIL-PRF-85045 for conformance inspection sample size.

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TABLE I. Qualification and conformance inspection.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/	Conformance inspection
I	Visual and mechanical inspection	3.4, 3.9, 3.10	4.7.2	Three samples, 0.5 km each 3/	A
	Attenuation rate	3.5.1	4.7.4.1	Three samples, 0.5 km each 4/	A
III	Temperature cycling	3.7.1	5/	Two samples, 0.5 km each 4/ (one on reel, one off)	C
	Temperature humidity cycling	3.7.3	4.7.6.3	Two samples, 0.5 km each 6/	C
	Storage temperature	3.7.4	4.7.6.4	Two samples, 0.5 km each 6/	--
	Low temperature flexibility (cold bend)	3.6.3	4.7.5.3	Two specimens, 8 m each 7/	B
	Cyclic flexing	3.6.4	4.7.5.4	Six specimens, 5 m each 8/ (two specimens at each temperature)	--
	Cable twist-bend	3.6.6	4.7.5.6	Six specimens, 5 m each 8/ (three specimens at each temperature)	--
	Impact	3.6.8	4.7.5.8	Two units, 5 m each 7/	--
	Temperature life (life aging)	5/	5/	Two specimens, 300 m each 7/	C
	Fungus resistance	3.8.4	4.8.4	Two specimens, 0.5 m each 7/	--
	Cable element removability	3.6.18	4.7.5.18	Two specimens, 0.5 m each 7/	C
IV	Thermal shock	3.7.2	4.7.6.2	One specimen, 0.49 km 4/ (on reel)	--
	Weathering	3.7.8	4.7.6.8	One specimen, 2 m 9/ and three material samples 10/	--
	Fluid immersion	3.7.9	4.7.6.9	One specimen, 2 m 9/ and three material samples 10/ for each specified fluid	C
	Paint susceptibility	3.7.15	4.7.6.15	Two specimens, 2 m each 9/	--
	Jacket self-adhesion or blocking	3.7.11	4.7.6.11	One specimen, 30 m 9/	--
	Shock	3.7.13	4.7.6.13	One specimen, 30 m 9/	--
	Hosing: low pressure	3.6.12.1	4.7.5.12.1	One specimen, 1.5 m 9/	C
	Hosing: hydrostatic	3.6.12.2	4.7.5.12.2	One specimen, 2 m 9/	C
	Radial compression (for application B only)	3.6.7	4.7.5.7	One specimen, 10 m 9/	--
	OFCC kink	5/	5/	Two specimens, 1 m each 9/	--
	Cross-link verification	5/	5/	Two specimens, 1 m each 9/	C

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TABLE I. Qualification and conformance inspection – Continued.

Group	Qualification inspection	Requirement paragraph	Test paragraph	Cable length 1/, 2/	Conformance inspection
V	Dripping	3.6.13	4.7.5.13	One specimen, 30 cm 9/	--
	Cable jacket tear strength	3.6.14	4.7.5.14	Five flat extruded jacket material strips 11/	C
	Cable jacket material tensile strength and elongation	3.6.15	4.7.5.15	Five specimens 12/	--
	Cable abrasion resistance	3.6.16	4.7.5.16	Four specimens, 2 m each 13/	C
	Cable shrinkage	3.6.17	4.7.5.17	Three specimens, 0.5 m each 14/	C
	Durability of identification	3.6.19	4.7.5.19	Three specimens, 2 m each 14/	--
	Stuffing tube compression	3.6.21	4.7.5.21	One specimen, 3 m 9/	--
	Flame extinguishing and smoke generation	3.7.12.2	4.7.6.12.2	One specimen, 50 m 9/	C
	Water absorption	3.7.14	4.7.6.14	Two specimens, extruded jacket material strips 12/	--
VI	Acid gas generation	3.8.1	4.8.1	One specimen, 1 m 15/	C
	Halogen content	3.8.2	4.8.2	One specimen, 1 m 15/	--
	Toxicity index	3.8.3	4.8.3	One specimen, 1 m 15/	C
	Smoke index	3.8.5	4.8.6	One specimen, 1 m 15/	C

NOTES:

1/ Tolerance on 0.5 km length is ±5 percent, provided that results are normalized to 1 km.
2/ Tolerance on shorter lengths is ±5 percent.
3/ The visual and mechanical inspection shall only be conducted on a 2-m section of each sample.
4/ The same samples as used in the visual and mechanical inspection shall be used.
5/ As stated under the applicable tests in this specification sheet.
6/ The same samples as used in the temperature cycling test shall be used.
7/ A specimen cut from each sample used in the temperature cycling test shall be used.
8/ Three specimens cut from each sample used in the temperature cycling test shall be used.
9/ Specimen cut from the specimen used in the thermal shock test shall be used.
10/ Three flat extruded jacket material strips as specified in 4.7.5.15 of MIL-PRF-85045 shall be used in this test.
11/ Flat extruded jacket material strips (i.e., strips of flat extruded material with the same composition as the cable jacket and of sufficient dimensions in which dumbbells can be cut) prepared as specified in 4.7.5.14 of MIL-PRF-85045 and obtained from flat extruded material that underwent the thermal shock test shall be used.
12/ Flat extruded jacket material as specified in 4.7.5.15 of MIL-PRF-85045 shall be used in this test.
13/ Specimen cut from the specimen used in the thermal shock test shall be used. Two specimens shall be used for scraping abrasion and two specimens shall be used in the cable-to-cable abrasion testing.
14/ Specimens cut from the specimen used in the thermal shock test shall be used.
15/ A specimen cut from one of the samples used in the temperature cycling test or specimen used in the thermal shock test shall be used.

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Qualification and conformance by similarity to previously qualified cable(s): Manufacturers who are qualified for M85045/17-01P (02P), M85045/22-01 (02), and M85045/18-01P (02P) or M85045/20-01M (02M) and whose multimode (single-mode) cable produced to this specification passes all tests and inspections specified in table II are qualified under this specification sheet for M85045/24-01 (02). This qualification by similarity is applicable only if the same 18-fiber cables (with the exception of cable jacket thickness) and same cable materials used in the previously qualified cables are used in the cable under test. Testing may be performed on a single length of cable, with a minimum length of 0.5 km.

TABLE II. Qualification and conformance by similarity (cable).

Group	Inspection	Qualification inspection 1/	Conformance inspection 2/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Temperature cycling	X	C
	Temperature humidity cycling	X	--
IV	Hosing: low pressure and hydrostatic	X	C
	Radial compression	X	--
	Cable twist-bending	X	C
	Impact (low temperature only)	X	C
V	Cable jacket tear strength	X	C
	Cable jacket material tensile strength and elongation	X	C
	Cross-link verification	X	C
VI	Flame extinguishing and smoke generation	X	C
	Acid gas generation	X	C

NOTES:

1/ Where manufacturers are qualified for M85045/17-01P (02P), M85045/22-01 (02), and M85045/18-01P (02P) or M85045/20-01M (02M) and are pursuing qualification for M85045/24-01 (02).

2/ Where manufacturers are qualified for M85045/17-01P (02P), M85045/22-01 (02), and M85045/18-01P (02P) or M85045/20-01M (02M) and are performing conformance testing for M85045/24-01 (02).

Qualification by similarity – fiber type: Manufacturers who are qualified under this specification sheet for multimode (M85045/24-01) fiber cable and whose single-mode fiber cable passes all tests and inspections specified in table III are qualified under this specification sheet for single-mode (M85045/24-02) fiber cable. This qualification by similarity is applicable if the only difference between the previously qualified cable and the cable under test is that the optical fiber has been changed from a multimode fiber to a single-mode fiber. Testing may be performed on either one or two lengths of cable, each with a minimum length of 0.5 km. Test order must be observed up to and including the storage temperature test. If only one cable length is used, the thermal shock test shall be performed after the storage temperature test.

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TABLE III. Qualification and conformance by similarity (fiber).

Group	Inspection	Qualification inspection M85045/24-02 1/	Conformance inspection M85045/24-02 2/
I	Visual and mechanical	X	A
	Attenuation rate	X	A
III	Temperature cycling	X	C
	Temperature humidity cycling	X	--
	Storage temperature	X	--
	Cyclic flexing	X	--
	Crush	X	--
	Cable twist-bending	X	C
	Impact (low temperature only)	X	C
IV	Thermal shock	X	--

NOTES:

1/ Where manufacturers are qualified for M85045/24-01 and are pursuing qualification for M85045/24-02.

2/ Where manufacturers are qualified for M85045/24-01 and M85045/24-02 and are performing conformance testing for M85045/24-02.

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CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6015-2022-002)

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